

Definition. Let $G_1, G_2, \dots, G_n$ be a finite collection of groups. The *external direct product* of $G_1, G_2, \dots, G_n$ is

Example 1. What is $\mathbb{Z}_2 \oplus D_3$?

Question 1. What is the order of $G_1 \oplus G_2$?

Question 2. What do you conjecture is the order of an element in $G_1 \oplus G_2$?

Theorem.

Question 3. How many elements of order 10 are there in $\mathbb{Z}_{20} \oplus \mathbb{Z}_{30}$?

Theorem. Let G and H be finite cyclic groups. Then $G \oplus H$ is cyclic if and only if

Proof:

Corollary.

Corollary.

Notation:

Theorem. Suppose s and t are relatively prime. Then

Exercise 1. What is the smallest positive integer k such that $x^k = e$ for all $x \in U(7 \cdot 17)$? Generalize to $U(pq)$ where p, q are distinct primes

Exercise 2. Determine the number of cyclic groups of order 15 in $\mathbb{Z}_{90} \oplus \mathbb{Z}_{36}$. What is a generator for each of these cyclic groups?

Exercise 3. How many elements of order 4 are there in $\text{Aut}(\mathbb{Z}_{20})$? Order 2?

Exercise 4. How many subgroups are there of $U(27)$

Exercise 5. Prove that $U(144) \approx U(140)$.

Exercise 6. Prove that $x^{100} = e$ for all $x \in U(1000)$.